

# Flutter Summery

## - What's Flutter?

Flutter is an open-source UI software development kit by Google, enables developers to build natively compiled apps for mobile, web, and desktop from a single codebase, and uses Dart programming language.

## - Why Flutter?

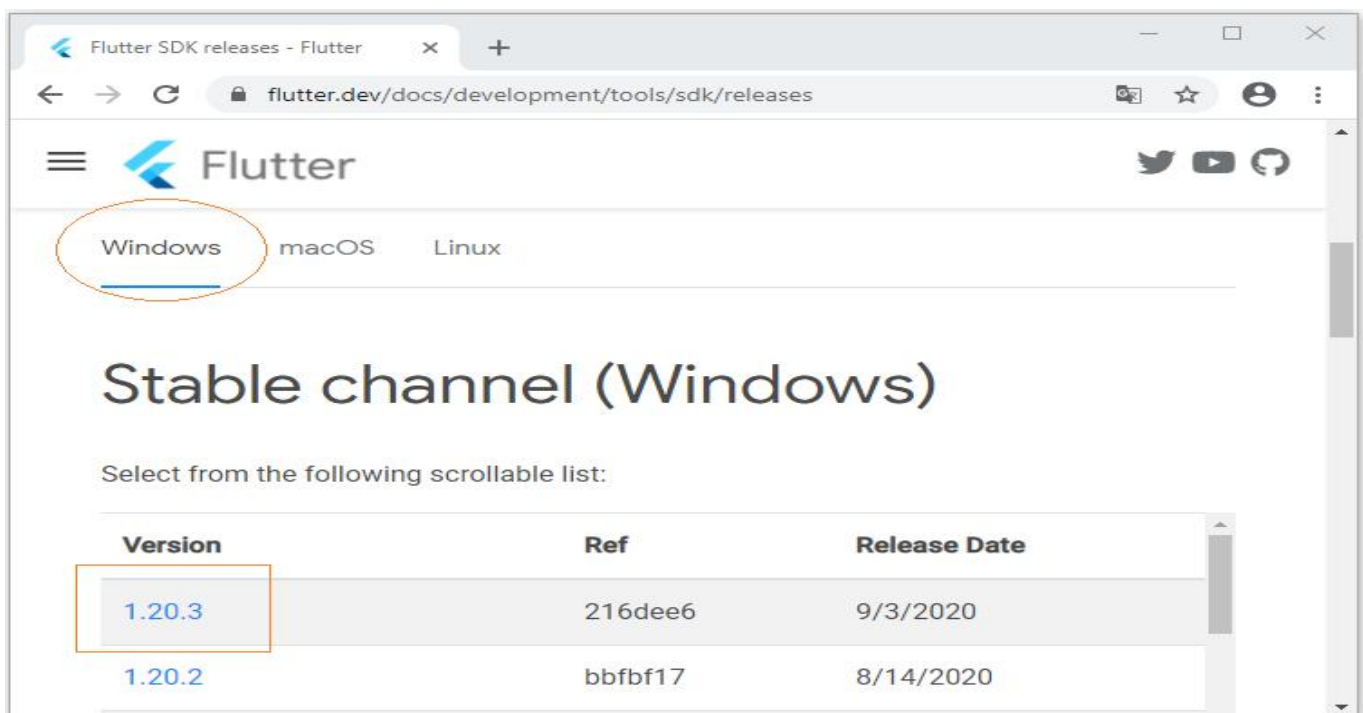
Choosing Flutter was based on many big benefits as:

- ✓ Single codebase reduces development time and cost.
- ✓ High-performance apps with native-like performance.
- ✓ Customizable UI with Flutter's wide range of widgets.
- ✓ Supports mobile, web, and desktop apps.
- ✓ Fast Development with 'Hot Reload'.

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## - Flutter Installation and Setup

Step 1: Download Flutter SDK: Flutter SDK contains: Dart SDK, Flutter Framework, and CLI tools, Open the official site then download the suitable SDK for your operating system, then Extract the archive to a suitable directory, for example: "C:\src\flutter"

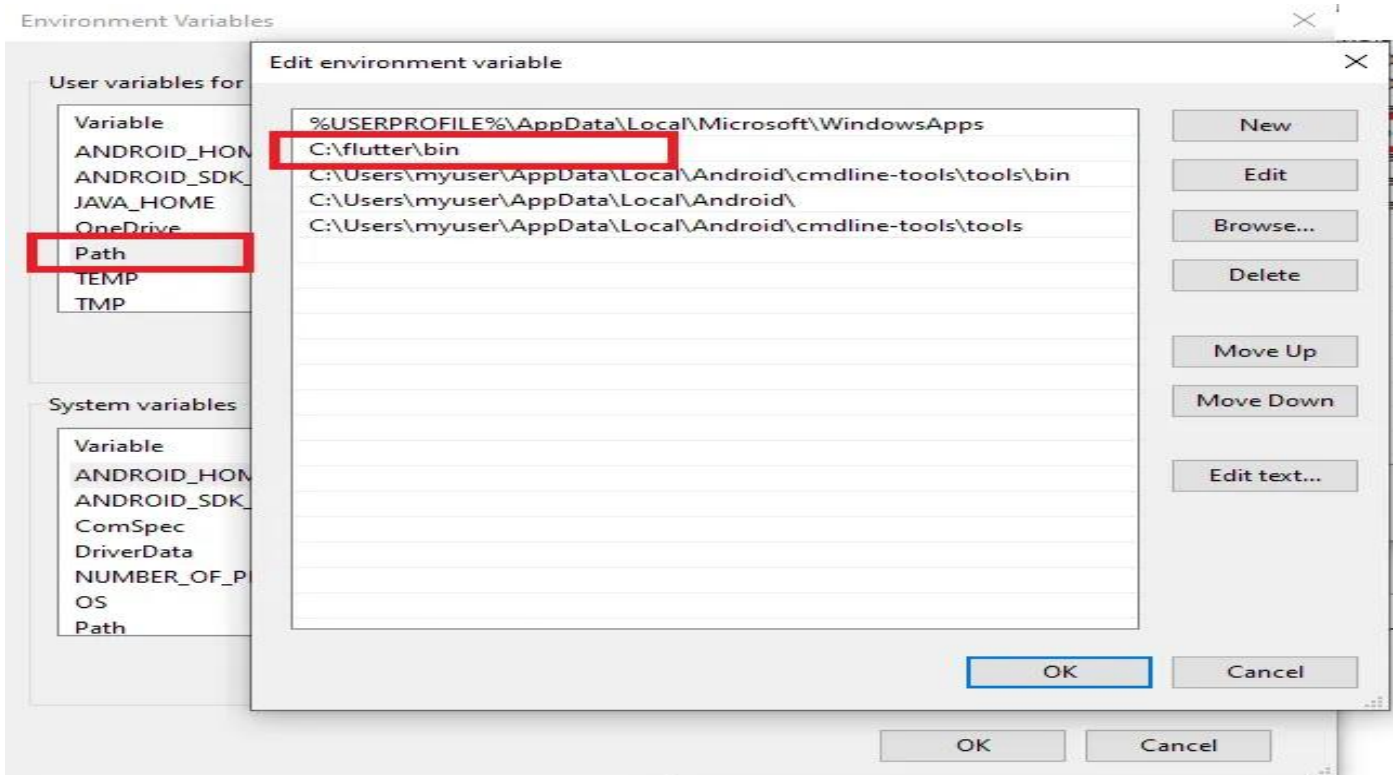


## Step 2: Add Flutter to PATH:

Add the “flutter/bin” directory to your system’s PATH environment variable.

On Windows:

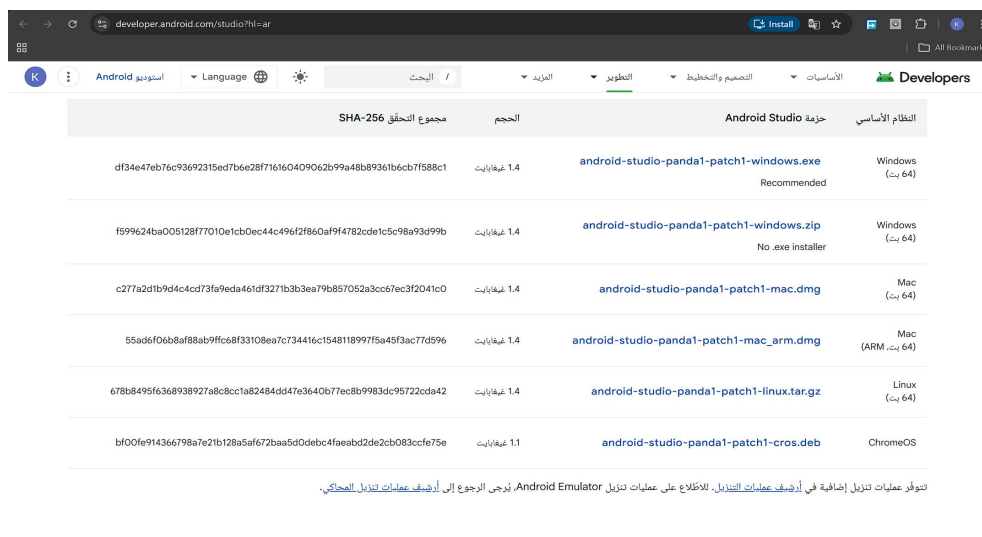
1. Open Environment Variables.
2. Edit the Path variable.
3. Add: add this path “C:\src\flutter\bin”



## Step 3: Install Android Studio

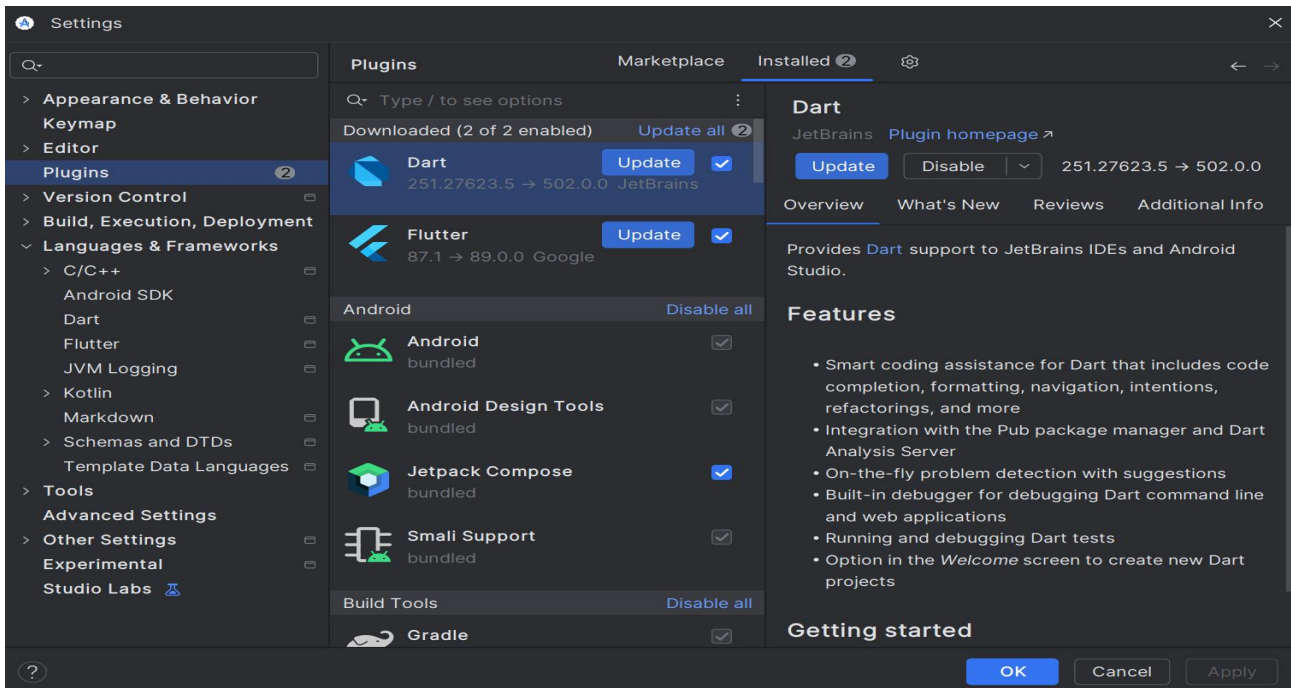
Android Studio is not used only as a code editor but it provides:

- Android SDK (Software Development Kit)
- Platform tools (ADB, build-tools)
- Android Emulator
- Device management tools



#### Step 4: Install Flutter and Dart Plugins

Open Android Studio and navigate to: Settings → Plugins, search for flutter and install it.



#### Step 5: Run Flutter Doctor

Open a terminal and execute: flutter doctor

This command checks: Flutter SDK installation, Dart configuration, Android SDK setup, Connected devices and Required licenses.

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\khaled_Aldumaini>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.35.3, on Microsoft Windows [Version 10.0.26200.7840], locale en-US)
[✓] Windows Version (11 Home 64-bit, 25H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.1.0-rc1)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.0.4)
[✓] Android Studio (version 2025.1.3)
[✓] VS Code (version unknown)
    X Unable to determine VS Code version.
[✓] Connected device (3 available)
[✓] Network resources

• No issues found!

C:\Users\khaled_Aldumaini>
```

## - Conclusion:

Flutter is a powerful and modern framework for cross-platform application development. It enables developers to build high-performance applications for mobile, web, and desktop from a single codebase.

Flutter is particularly well-suited for startups, as it reduces development costs and shortens time-to-market. Its rich widget system and strong community support make it an efficient choice for rapid product iteration.

Overall, Flutter simplifies and accelerates the development process, making it an excellent solution for teams aiming to deliver scalable and visually appealing applications quickly.